

Camillo Maria Caruso

AI RESEARCH SCIENTIST — MULTIMODAL & ROBUST DEEP LEARNING

Rome, Italy

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Summary

Ph.D. researcher specialized in multimodal and generative AI for healthcare. Experienced in designing and deploying transformer-based architectures robust to missing data across heterogeneous modalities (images, tabular, text). Skilled in PyTorch, Docker, HPC workflows (SLURM), and reproducible ML pipelines. Strong academic output with practical R&D experience.

Experience

University Campus Bio-Medico of Rome — Unit of Artificial Intelligence & Computer Systems (ArCo)

Rome, Italy

RESEARCH FELLOW

Nov 2024 – Present

- Developed multimodal and generative AI solutions for diagnosis, prognosis, and treatment support.
- Designed transformer architectures robust to missing data for tabular and imaging modalities.
- Led integration of PyTorch pipelines on HPC clusters using Docker and SLURM.
- Mentored Ph.D. students and contributed to EU project proposal preparation.

University Campus Bio-Medico of Rome

Rome, Italy

COLLABORATING PROFESSOR — MACHINE LEARNING & STATISTICS

2022 – 2025

- Lecturer for *Machine Learning & Big Data Analysis* (MSc Biomedical Engineering).
- Lecturer for *Statistics* (Medicine and Surgery).

Education

University Campus Bio-Medico of Rome

Rome, Italy

PH.D. IN ARTIFICIAL INTELLIGENCE (HEALTH & LIFE SCIENCES)

2021 – Present

- Thesis: *Resilient Multimodal Learning with Incomplete Information in Biomedical Applications*.
- Research exchange: Umeå University, Sweden (Biomedical Engineering Dept.).

University Campus Bio-Medico of Rome

Rome, Italy

M.Sc. IN BIOMEDICAL ENGINEERING — 110/110 *cum laude*

2018 – 2021

- Thesis: *Analysis and Classification of Relevant Interventions to the National Fire Corps*.
- Focus: Machine Learning, Big Data, Image Processing, IoT Systems.

Skills

Programming Python (PyTorch, NumPy, Pandas, scikit-learn), MATLAB, C++

Tools Docker, Git, Linux/Unix, HPC (SLURM), LaTeX

Languages Italian (native), English (C1)

Selected Publications

- MARIA: A Multimodal Transformer Model for Incomplete Healthcare Data.** *Computers in Biology and Medicine*, 2025. doi:10.1016/j.compbimed.2025.110843
- A Deep Learning Approach for Survival Prediction in Lung Cancer with Missing Values.** *Computer Methods and Programs in Biomedicine*, 2024. doi:10.1016/j.cmpb.2024.108308
- A Multimodal Ensemble Driven by Multi-objective Optimization for NSCLC Prognosis.** *Journal of Imaging*, 2022. doi:10.3390/jimaging8110298

Research Projects

- LUCIA: Lung Cancer Holistic Care via Interactive AI Agents.** EU Pathfinder Challenge Proposal (submitted).
- MedCoDi+ (2025–2026):** Multimodal Generative AI for Medical Image and Report Synthesis.
- FAIR (2023–2026):** Resilient Multimodal AI Systems.
- AIDA (2023–2026):** Explainable Multimodal Deep Learning for Oncology.
- PICTURE (2023–2026):** AI-driven Prediction of Therapy Response in NSCLC.
- IDEA (2023–2026):** Digital Twin for Lung Cancer Care.